



Mangroves vs Waves

My name _____

Class _____

Date _____



How can nature protect our coast from waves?

Waves can wash beaches away. Mangrove trees grow along the coast near Bangkok. Their roots stand in the water. Do they protect the land?



My kit tick when you have it

- ☐ A deep plastic tray
- ☐ Damp sand
- ☐ A jug of water
- ☐ Towels for spills
- ☐ 10 sticks for mangrove roots
- ☐ A ruler to make waves
- ☐ 15 sticks



Stay safe

- Wipe up spills so nobody slips.
- Keep sand away from your eyes.
- Wash your hands after touching sand.

START HERE: BUILD



Build your beach

1 · DO IT: TICK EACH STEP

- 1 ☐ Pile damp sand at one end of the tray.
- 2 ☐ Pat it into a smooth slope, like a beach.
- 3 ☐ Pour water slowly into the other end.
- 4 ☐ Stop when the water just touches the sand.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	Is your beach ready?
My beach	☒ Smooth slope and water ☐ The sand keeps sliding

Draw your beach

WORDS TO USE

sand beach slope sea water coast

TEST 1



Do roots help?

? Will "mangrove roots" stop the sand washing away?

1 · MY GUESS: CIRCLE ONE

 With no roots

 With roots

 Both the same

How sure am I?   

2 · DO IT: TICK EACH STEP

- ☐  Test 1: build a smooth sand slope.
- ☐  Leave the beach bare, with no sticks.
- ☐  Push a ruler in the water to make 10 waves.
- ☐  Look at the beach. Tap how much washed away.
- ☐  Test 2: build a smooth sand slope.
- ☐  Push 10 sticks into the sand at the water.
- ☐  Push a ruler in the water to make 10 waves.
- ☐  Look at the beach. Tap how much washed away.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How much sand washed away?
 No roots	 None  A little  Lots
 With roots	 None  A little  Lots

 Draw the beach after the waves: No roots

 Draw the beach after the waves: With roots

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right

☐ Partly right

☐ Something different happened

☐ Not sure yet

WORDS TO USE

waves roots mangrove sand washed protect erosion

TEST 2 Thin or thick forest?

? Does a thicker mangrove forest protect the beach better?

1 · MY GUESS: CIRCLE ONE

 3 sticks

 15 sticks

 No difference

How sure am I?   

2 · DO IT: TICK EACH STEP

- ☐  Test 1: build a smooth sand slope.
- ☐  Push 3 sticks into the sand at the water.

- 3 ☐ 🌊 Push a ruler in the water to make 10 waves.
- 4 ☐ 👁️ Look at the beach. Tap how much washed away.
- 5 ☐ 🏖️ Test 2: build a smooth sand slope.
- 6 ☐ 🌳 Push 15 sticks into the sand at the water.
- 7 ☐ 🌊 Push a ruler in the water to make 10 waves.
- 8 ☐ 👁️ Look at the beach. Tap how much washed away.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How much sand washed away?
🪵 3 sticks	🌊 None 🖐️ A little 🌊 Lots
🌳 15 sticks	🌊 None 🖐️ A little 🌊 Lots

🪵 Draw the beach after the waves: 3 sticks

🪵 Draw the beach after the waves: 15 sticks

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right
 ☐ 🌐 Partly right
 ☐ 🔄 Something different happened
 ☐ 🤔 Not sure yet

WORDS TO USE

thick
 thin
 forest
 roots
 more
 less
 protect

🎨 **Art corner: Mangrove home**

Draw a mangrove forest. Show the roots under the water, and the crabs, fish and birds that live there.

➗ **Maths corner: Stick maths**

Draw 3 sticks and 15 sticks. Which is more?

📖 **New words**

Coast: Where the land meets the sea.
 Erosion: When water or wind carries land away.
 Protect: To keep something safe from harm.

★ **What I can do now** colour the stars

I can make a model beach and make waves the same way each time.



I can see how roots change what waves do to sand.



I can explain why mangroves are important.

